Suyog Mainali

Williamsburg, VA • +1 (757) 206–8339 • mainalisuyog0@gmail.com
linkedin.com/in/suyog41 • github.com/suyog12 • suyog12.github.io/suyog-mainali-portfolio

Professional Summary

Senior Data Engineer with 5+ years of experience designing and owning large-scale batch and streaming data systems using Apache Spark (PySpark), Kafka, Airflow, and dbt. Proven track record building fault-tolerant pipelines processing 10TB+ daily and powering real-time data services and ML workloads on AWS (EMR, S3, Glue, Lambda, Redshift).

Core Skills

Big Data & Lakehouse: Apache Spark (PySpark), Azure Databricks, Spark SQL, Delta Lake, ETL/ELT pipelines, batch & streaming systems, Kafka, Airflow, dbt
Cloud & Backend: AWS (EMR, S3, Glue, Lambda, Step Functions), Azure ADLS Gen2, Snowflake, Redshift, REST APIs
Databases & Storage: PostgreSQL, MySQL, MongoDB, Elasticsearch, Parquet, JSON
DevOps & Platform: Docker, Kubernetes, CI/CD pipelines, Git, JIRA, Linux, monitoring, logging, SLA ownership
Analytics & Visualization: Tableau, Power BI
Healthcare Domain: Claims and EHR integration, HIPAA-compliant data pipelines, PHI governance

Professional Experience

Zakipoint Health *Senior Data Engineer | July 2020 – July 2025*

- Architected the company's flagship real-time cost prediction engine, processing 10 TB+ daily across claims and provider data to deliver sub-second pre-visit out-of-pocket estimates to members.
- Took ownership of unstable core data pipelines and rebuilt PySpark and SQL workflows on AWS Glue, cutting recurring processing failures by 60%+ and stabilizing downstream analytics and ML systems.
- Built executive Tableau dashboards, eliminating 2-day manual reporting and supporting data-driven decisions.
- Enabled production AI models by designing feature pipelines from claims and clinical data, reducing ML data preparation time by 40% for risk scoring and care gap detection.
- Owned analytics reliability across Snowflake, Redshift, Tableau, and CI/CD workflows, maintaining 99.8% uptime under HIPAA and SOC 2 requirements.

Om Hospital *Data Engineer | April 2018 – July 2020*

- Led migration from paper records and legacy systems to DHIS2 for a hospital handling 250K+ annual visits, preserving multi-year historical patient data.
- Built the hospital's first centralized data warehouse integrating visits, diagnoses, labs, and operations, enabling organization-wide clinical reporting for the first time.
- Implemented a digital visit recording and telemedicine system in Python, supporting engagement with 10K+ patients per month and enabling longitudinal patient histories across departments.
- Trained 50+ doctors, nurses, and admin staff through workshops and documentation, driving adoption and consistent use of new digital workflows.

Projects

Healthcare Churn & Specialty Analytics (HIPAA-Restricted)
Built end-to-end analytics pipelines on claims and EHR-derived data supporting churn and specialty utilization modeling; performed data cleaning, comorbidity and severity score computation in Python, and developed logistic regression, random forest, and gradient-boosted models in R with robust validation under HIPAA constraints.
Technologies: Python, R, SQL, healthcare feature engineering, data validation

Amazon Reviews ETL Pipeline GitHub Repository
Built production-grade ETL pipeline ingesting and cleaning 500K+ records with retry, resume, and validation logic; stored analytics-ready datasets in PostgreSQL for downstream analysis and reporting.
Technologies: Python, MySQL, ETL design, data quality checks

Professional Certifications

Mastering Azure Databricks for Data Engineers – Packt/Coursera (Certificate)
Google Data Analytics – Google/Coursera (Certificate)
Data Analysis with R – Duke University/Coursera (Certificate)

Education

William & Mary, Mason School of Business | MS Business Analytics — Expected May 2026
Vellore Institute of Technology | B.Tech Computer Science Engineering — 2018 – 2022

Work Authorization: F-1 STEM OPT (3 years U.S. work eligibility; sponsorship required thereafter)